How This Week Works

Read first

This week your team is a real **engineering firm**. You get hired to solve a real problem, design a product that fixes it, and pitch it Friday to a review panel. **AI builds your image and slides – your job is to know your design cold.**

1 The week, day by day

1. **Mon** – form your firm, pick your problem (page 1).
2. **Tue** – define it like an engineer (page 2).
3. **Wed** – design the solution + generate your product image (page 3).
4. **Thu** – build the pitch, drill the tough questions (page 4).
5. **Fri** – Design Review: pitch + defend (page 5).

2 Your 4 roles

Lead Engineer
Runs design decisions, keeps the firm on task, calls the vote.

Design Lead
Owns the sketches and the AI product image.

Researcher
Proves the problem is real – who has it, evidence.

Presenter
Runs Friday's pitch (everyone speaks, they lead).

3 House rules

- Everyone contributes – no passengers.
- All work goes in this packet.
- Disagree? The Lead Engineer calls a 30-second vote.
- Stuck? Raise one **"firm question,"** not four hands.

Page 1 – Start Your Firm

Mon · Day 6

Firm: ______________________ **Members:** ______________________

1 Firm name

2 Members + roles

NAME	ROLE

3 Choose ONE problem

- ☐ Backpacks wreck your shoulders and back.
- ☐ Phones die before the school day is over.
- ☐ Water bottles leak inside a backpack.
- ☐ Nowhere on a desk for your phone, pencil, and earbuds.
- ☐ Doors slam loud or won't stay open when your hands are full.
- ☐ Earbuds and charger cables tangle and disappear.
- ☐ Lockers are a mess – stuff falls out every time.
- ☐ Hard to carry a lunch tray + a drink + your bag at once.
- ☐ Our own (teacher-approved): ______________________

4 Why it matters

Who has this problem, and how often?

Page 2 – Define It Like an Engineer

Firm: ____________________ **Members:** ________________________________

A real definition has 4 parts
Who has it · evidence it's real · requirements a fix MUST meet · how you'll know it worked.

1 Who has it (be specific)

2 Evidence it's real (what you've seen or counted)

3 Requirements – 3 to 5, each checkable (a number or yes/no)

At least one must be a **constraint** (cost, size, or materials).

4 Success =

Not a requirement: "it should be cool / nice / good." Rewrite anything like that as something you can measure.

Page 3 – Design It + Render It

Wed · Day 8

Firm: ____________________ **Members:** ____________________

1 Sketch 3 different solutions

idea 1 – label the key part

idea 2 – label the key part

idea 3 – label the key part

2 Chosen design + why + one trade-off

3 Product-image prompt (fill the blanks, paste into your image AI)

"A clean product photo of a _____ that helps _____ by _____. Made of _____, about _____ big. Realistic, plain white background, studio lighting."

- [ ] Product image generated and added to Google Classroom.

Page 4 – Build & Drill the Pitch

Thu · Day 9

Firm: ______________________ **Members:** ______________________

1 Your 5-beat pitch

BEAT	YOUR ONE LINE
Problem	
Evidence	
Solution (+ image)	
How it works	
Impact	

2 Drill the defense – who owns each answer?

THE PANEL WILL ASK…	WHO ANSWERS
Why this and not another idea?	
How does it actually work?	
What would it cost / what's it made of?	
Who exactly is this for?	

- [ ] 5-slide deck posted to Classroom
- [ ] One full practice run done.

Page 5 – Design Review

Fri · Day 10

Firm: ______________________ **Members:** ______________________

3-minute pitch + 2-minute defense in front of the panel. Everyone on the firm speaks.

1 Rubric – 25 points

What we score	Points
Problem is clear and real	/5
Solution fits the requirements	/5
How it works makes sense	/5
Defense – answered the questions	/5
Teamwork – everyone contributed	/5
Total	/25

2 Peer judge (a firm that isn't yours)

Firm judged: ______________________ "What I liked is _____. My question is _____."

3 Reflection

One thing your firm did well, and one thing you'd change next time.